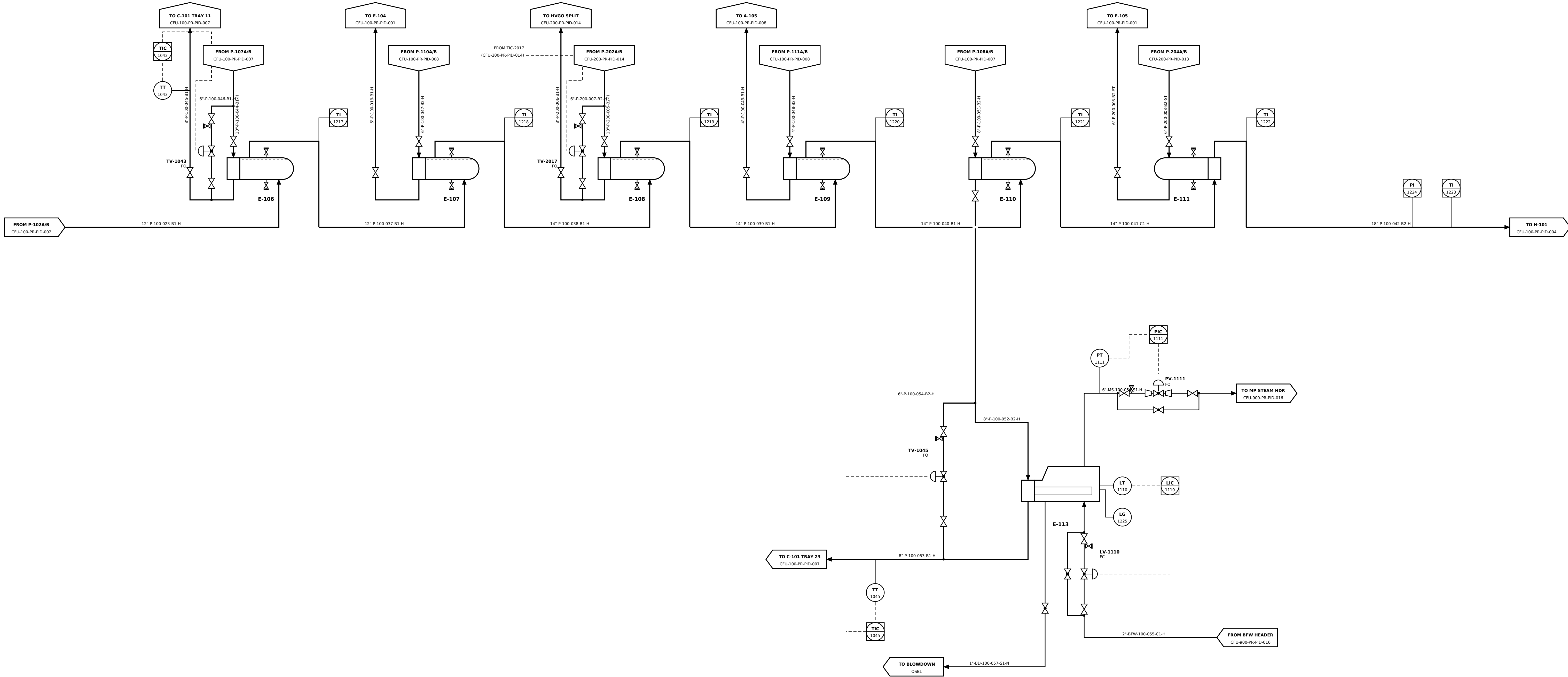


E-106 Crude / Middle (kero) pumparound 1001 m2 total; 25 x 1P shells DESIGN 30 / 20 barg / 270 C; CA 3 mm DUTY 5.83 MW SHELL: Crude / TUBE: Middle (kero) pumparound MATL CS shell / 5Cr-1/2Mo tubes	E-107 Crude / Diesel product 324 m2 total; 15 x 1P shells DESIGN 30 / 20 barg / 325 C; CA 6 mm DUTY 12.85 MW SHELL: Crude / TUBE: Diesel product MATL CS shell / 5Cr-1/2Mo tubes	E-108 Crude / HVGO pumparound/product 2196 m2 total; 25 x 2P shells DESIGN 30 / 20 barg / 345 C; CA 6 mm DUTY 21.14 MW SHELL: Crude / TUBE: HVGO pumparound/product MATL 1.25Cr shell / 9Cr-1Mo tubes	E-109 Crude / AGO product 263 m2 total; 15 x 1P shells DESIGN 30 / 20 barg / 365 C; CA 6 mm DUTY 3.11 MW SHELL: Crude / TUBE: AGO product MATL 1.25Cr shell / 9Cr-1Mo tubes	E-110 Crude / Bottom (diesel) pumparound 886 m2 total; 15 x 2P shells DESIGN 30 / 20 barg / 345 C; CA 6 mm DUTY 7.46 MW SHELL: Crude / TUBE: Bottom (diesel) pumparound MATL 1.25Cr shell / 9Cr-1Mo tubes	E-111 Crude / Vacuum residue 1094 m2 total; 15 x 2P shells DESIGN 30 / 20 barg / 395 C; CA 6 mm DUTY 7.58 MW SHELL: Vacuum residue / TUBE: Crude MATL 1.25Cr shell / 9Cr-1Mo tubes	E-113 BPA / MP steam generator (kettle) 223 m2; 1 shell(s) DESIGN 13 / 20 barg / 290 C; CA 3 mm DUTY 4.91 MW SHELL: BFW / MP steam / TUBE: Bottom pumparound MATL CS
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- NOTES**
- Hot train crude-side design pressure per P-102 shut-off (see line list); exchanger rating to be confirmed.
 - E-113 kettle MP steam generator: 3-element level control simplified to LIC-1110 at FEED.
 - TV-2017 controlled from TIC-2017 (HVGO pumparound return temperature) on CFU-200-PR-PID-014.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
HOT PREHEAT TRAIN E-106 TO E-111 & E-113			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	4 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-003			A